

Chapter 8

Differential Rules(Part 2)

§ Section: 3.4-3.5

8.1 The Chain Rule

The Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composite $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$= f'(g(x)) \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Idea of the chain rules

Let $u = g(x)$, $y = f(u) = f(g(x))$. If the variable x changes to $x + \Delta x$, then the change of u and y are

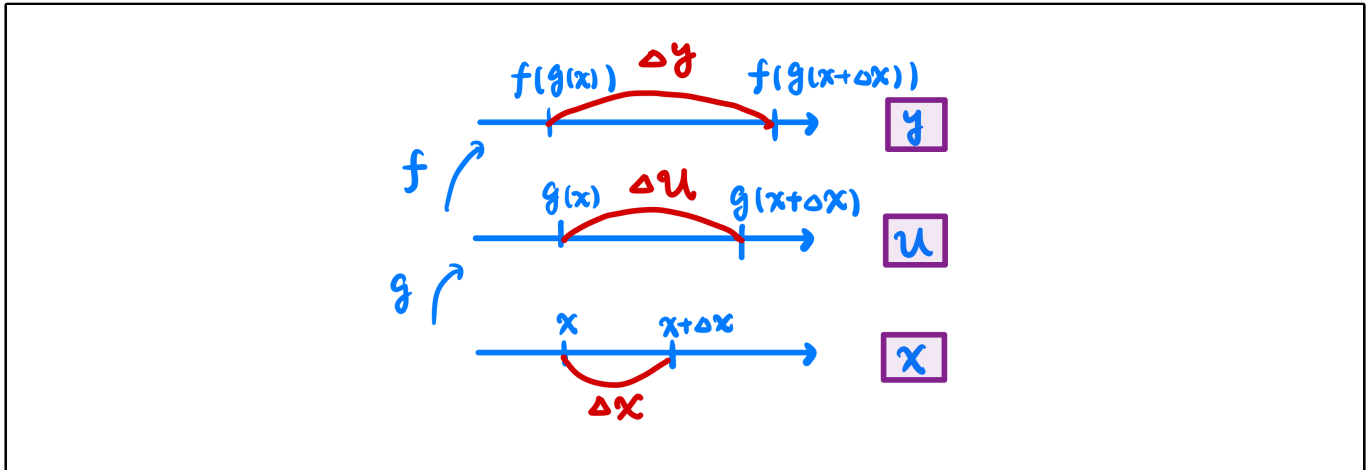
$$\Delta u = g(x + \Delta x) - g(x) \text{ and } \Delta y = f(g(x + \Delta x)) - f(g(x))$$

Hence,

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$$

that is,

$$\frac{f(g(x + \Delta x)) - f(g(x))}{\Delta x} = \frac{f(g(x + \Delta x)) - f(g(x))}{g(x + \Delta x) - g(x)} \cdot \frac{g(x + \Delta x) - g(x)}{\Delta x}$$

**Example 1.1**

A table of values for f , g , f' and g' are given.

(a) $F(x) = f(g(x))$. Find $F'(1)$.

(b) $G(x) = g(f(x))$. Find $G'(2)$.

(c) $H(x) = g(g(x))$. Find $H'(3)$.

(d) $h(x) = (g(x))^2$. Find $h'(3)$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	3	6	3
2	3	1	7	5
3	1	2	4	-1

sol: a) $F'(1) = f'(g(1)) \cdot g'(1) = f'(3) \cdot g'(1) = 4 \cdot 3 = 12.$

b) $G'(2) = g'(f(2)) \cdot f'(2)$
 $= g'(3) \cdot f'(2) = (-1) \cdot 7 = -7.$

Example 1.2

Suppose that $L(x)$ is a function such that $L'(x) = \frac{1}{x}$ for $x > 0$. Find

(a) $\frac{d}{dx} L(c \cdot x)$, where $c \in \mathbb{R}$, $c > 0$.

(b) $\frac{d}{dx} L(x^n)$, where $n \in \mathbb{R}$.

(c) $\frac{d}{dx} L(f(x) \cdot g(x))$, where $f(x)$ and $g(x)$ are positive differentiable functions.

sol: a) $\frac{d}{dx} L(c \cdot x) = L'(cx) \cdot (cx)' = \frac{1}{cx} \cdot c = \frac{1}{x}$

b) $\frac{d}{dx} L(x^n) = L'(x^n) \cdot (x^n)' = \frac{1}{x^n} \cdot nx^{n-1} = \frac{n}{x}$

c) $\frac{d}{dx} L(f(x) \cdot g(x)) = L'(f(x)g(x)) \cdot (f(x) \cdot g(x))'$

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$$= \frac{1}{f(x)g(x)} \cdot (f'g + fg') = \frac{f'(x)}{f(x)} + \frac{g'(x)}{g(x)}$$

Example 1.3

$f(x) = \sin(x^2 + 1)$. Compute $f'(x)$.

$$= \sin(x^2 + 1)$$

$$\begin{aligned} f'(x) &= \cos(x^2 + 1) \cdot 2x \\ &= 2x \cos(x^2 + 1) \end{aligned}$$

The Power Rules Combined with the Chain Rule

If n is any real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = u n^{n-1} \frac{du}{dx}$$

Alternatively,

$$\frac{d}{dx} (g(x))^n = n (g(x))^{n-1} \cdot g'(x)$$

$$\frac{d}{dx} [g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$$

Example 1.4

$f(x) = (\sin x)^{\sqrt{2}}$. Compute $f'(x)$.

$$= (\sin x)^{\sqrt{2}}$$

$$\begin{aligned} f'(x) &= \sqrt{2} (\sin x)^{\sqrt{2}-1} \cdot \cos x \\ &= \sqrt{2} \cdot \cos x (\sin x)^{\sqrt{2}-1} \end{aligned}$$

Example 1.5

$$\begin{aligned}
 \text{Find } \frac{d}{dx}(x^2 + e^x)^{\frac{1}{3}} &= \frac{d}{dx} (x^2 + e^x)^{\frac{1}{3}} \\
 &= \frac{1}{3} (x^2 + e^x)^{-\frac{2}{3}} \cdot (2x + e^x) \\
 &= \frac{1}{3} (2x + e^x) (x^2 + e^x)^{-\frac{2}{3}}.
 \end{aligned}$$

Example 1.6

$$\begin{aligned}
 \text{Find } \frac{d}{dx} \frac{(3x-2)^4}{(x^2+x+1)^3} &= \frac{d}{dx} (3x-2)^4 \cdot (x^2+x+1)^{-3} \\
 &= 4(3x-2)^3 \cdot 3 \cdot (x^2+x+1)^{-3} + (3x-2)^4 \cdot (-3)(x^2+x+1)^{-4} \cdot (2x+1) \\
 &= 12 \frac{(3x-2)^3}{(x^2+x+1)^3} - 3 \frac{(3x-2)^4 (2x+1)}{(x^2+x+1)^4}
 \end{aligned}$$

Example 1.7

$$\begin{aligned}
 \text{Find } \frac{d}{dx} f(g(h(x))) &= f'(g(h(x))) \cdot (g(h(x)))' \\
 &= f'(g(h(x))) \cdot \underline{g'(h(x)) \cdot h'(x)}
 \end{aligned}$$

Example 1.8

$$\begin{aligned}
 \text{Find } \frac{d}{dx} (\sec(\sin x))^2 &= \frac{d}{dx} (\sec(\sin x))^2 = 2(\sec(\sin x)) \cdot (\sec(\sin x))' \\
 &= 2 \sec(\sin x) \cdot \underline{\sec(\sin x) \tan(\sin x) \cdot \cos x}
 \end{aligned}$$

Example 1.9

$F(x) = f(g(x))$. Find $F''(x)$.

$$F'(x) = f'(g(x)) \cdot g'(x)$$

$$\begin{aligned} F''(x) &= \left(f''(g(x)) \cdot g'(x) \right) \cdot g'(x) + f'(g(x)) \cdot g''(x) \\ &= f''(g(x)) \cdot (g'(x))^2 + f'(g(x)) \cdot g''(x) \end{aligned}$$

Example 1.10

$$\begin{aligned} \frac{d}{dx}(f(ax)) &= \frac{d}{dx} f(ax) = f'(ax) \cdot a \\ &= a f'(ax) \end{aligned}$$

Derivatives of General Exponential Functions

$$\frac{d}{dx}(a^x) = a^x \ln a \text{ for all } a > 0.$$

Example 1.11

$$\begin{aligned} \frac{d}{dx}(a^x) &= \frac{d}{dx} (e^{\ln a \cdot x}) = e^{\ln a \cdot x} \cdot \ln a \\ &= a^x \cdot \ln a \end{aligned}$$

Example 1.12

$$\begin{aligned} \frac{d}{dx}(3^{\sqrt{x}}) &= (3^{\sqrt{x}})' \\ &= \ln 3 \cdot 3^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{\ln 3}{2\sqrt{x}} \cdot 3^{\sqrt{x}} \end{aligned}$$

Example 1.13

$$\frac{d}{dx}(e^{2^x} + 2^{x^e} + x^{2^e})$$

$$\frac{d}{dx}(e^{2^x}) = e^{2^x} \cdot (2^x \cdot \ln 2)$$

$$(2^{x^e})' = \ln 2 \cdot 2^{x^e} \cdot (e \cdot x^{e-1}) = \ln 2 \cdot e \cdot x^{e-1} \cdot 2^{x^e}$$

$$(x^{2^e})' = 2^e \cdot x^{2^e-1}$$

Example 1.14

$$\frac{d}{dx}(4^{3^{2^x}}) = (4^{3^{2^x}})' = \ln 4 \cdot 4^{3^{2^x}} \cdot (3^{2^x})'$$

$$= \ln 4 \cdot 4^{3^{2^x}} \cdot (\ln 3 \cdot 3^{2^x} \cdot \ln 2 \cdot 2^x)$$

$$= \ln 2 \cdot \ln 3 \cdot \ln 4 \cdot 2^x \cdot 3^{2^x} \cdot 4^{3^{2^x}}$$

Example 1.15

Find $\lim_{x \rightarrow 0} \frac{3^{\sin 2x} - 1}{x} = f'(0)$, where $f(x) = 3^{\sin 2x}$

$$f'(x) = (3^{\sin 2x})' = \ln 3 \cdot 3^{\sin 2x} \cdot (\sin 2x)'$$

$$= \ln 3 \cdot 3^{\sin 2x} \cdot (\cos(2x) \cdot 2)$$

$$f'(0) = 2 \cdot \ln 3$$

Example 1.16

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & , \text{ for } x \neq 0 \\ 0 & , \text{ for } x = 0 \end{cases}.$$

- (a) Is f continuous at $x = 0$?
 (b) Compute $f'(0)$ and $f'(x)$, ~~for $x \neq 0$.~~
 (c) Is $f'(x)$ continuous at $x = 0$?

a) $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0 = f(0)$
 By the Squeeze Thm

$\Rightarrow f(x)$ is conti at $x=0$.

b) For $x \neq 0$, $f'(x) = 2x \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot \left(\frac{-1}{x^2}\right)$
 $= 2x \cdot \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$

$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$

$f'(0) = \frac{d}{dx}(f(0)) = 0 \times$

c) $\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} 2x \cdot \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) \quad \text{DNE}$

$\Rightarrow f'(x)$ is NOT conti at $x=0$.